Influences of data processing techniques on the interpretation of atmospheric spectra from JWST

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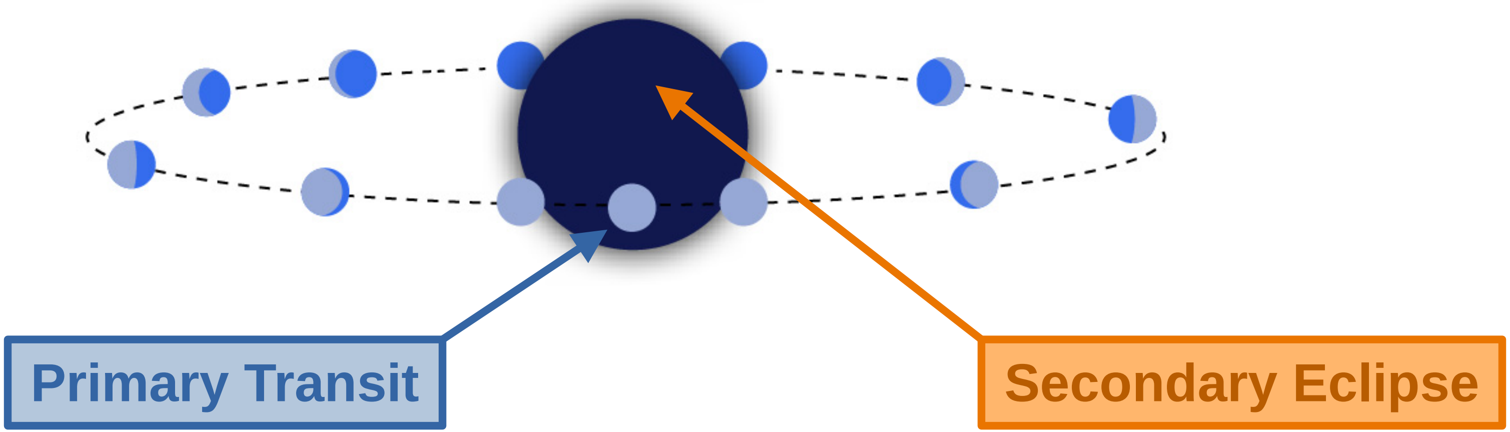
Image credit: NASA, ESA, CSA, Joseph Olmsted (STScI) [1]

Transit Spectroscopy

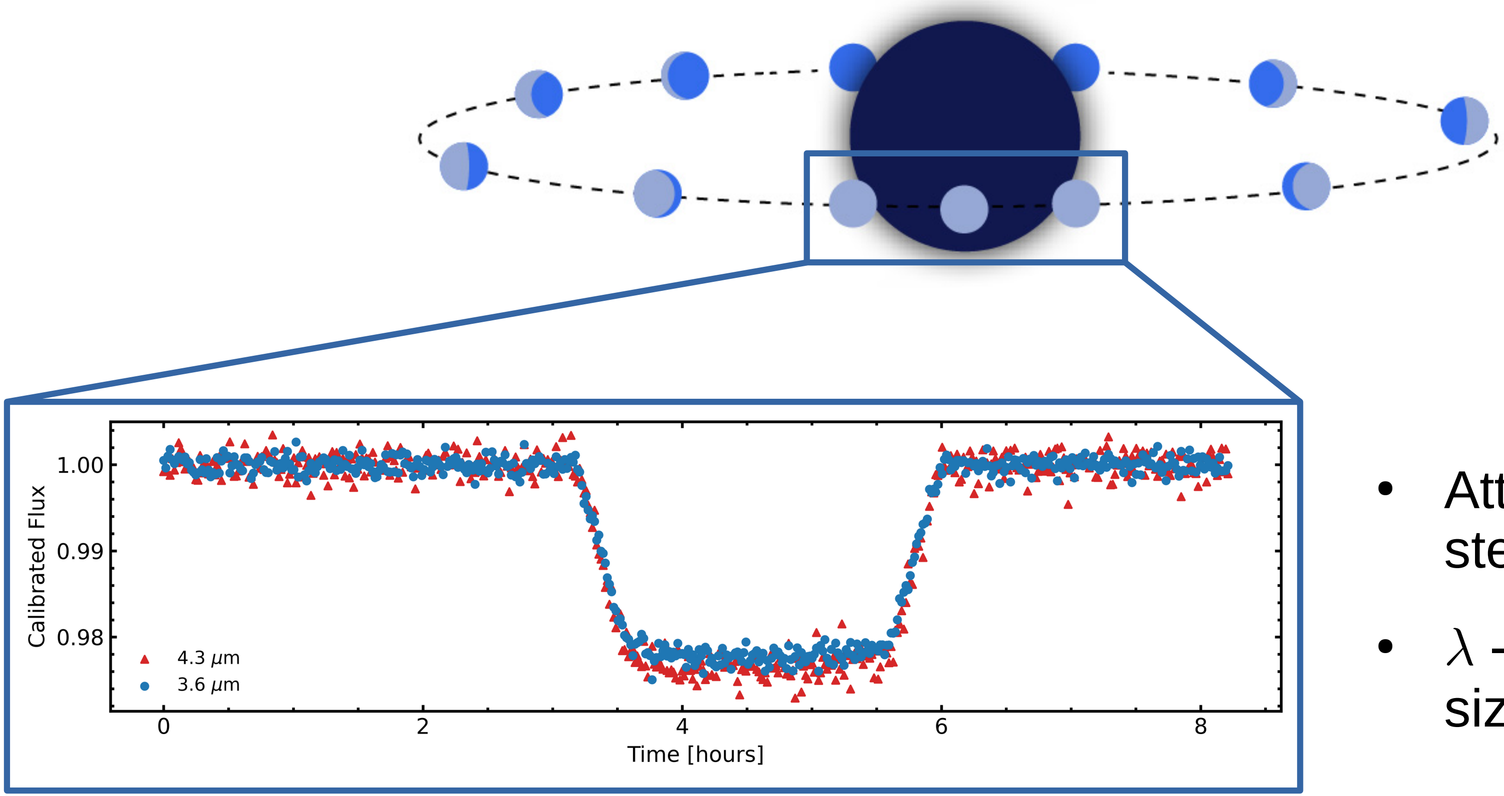
Image credit: ESA [2]



- High-cadence time-series observations with JWST
- Capture stellar flux in and out of transit event
  - Information dependent on orbital phase (Transmission / Emission)

Observing Primary Transits

Image credit: ESA [2]

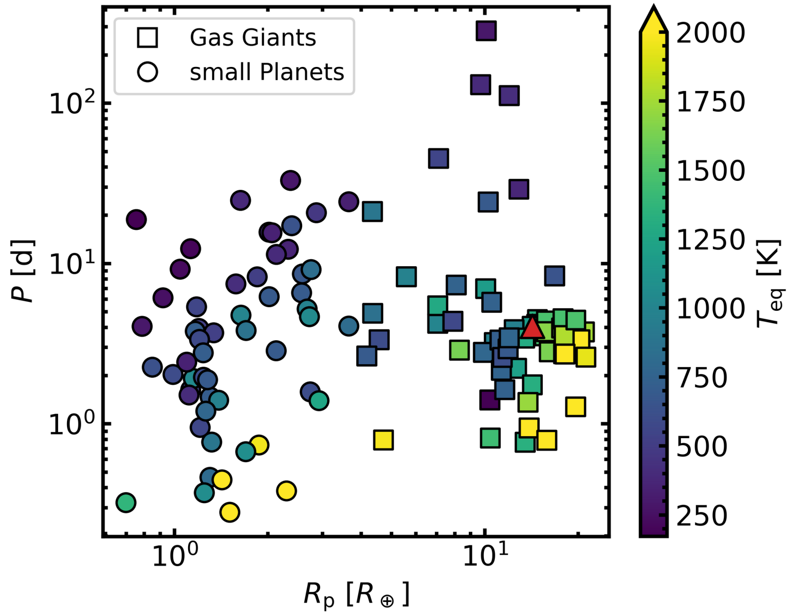


- Attenuation of stellar flux
- λ - dependent size of planet

Observational Data

JWST NIRSpec PRISM observation of WASP-39 b

- 0.6 – 5.3 microns (partial saturation)
  - effective coverage: λ > 2.2 μm
- Hot Jupiter orbiting a solar-like star
  - T_{eq} = 1100 K, SpT = G8V



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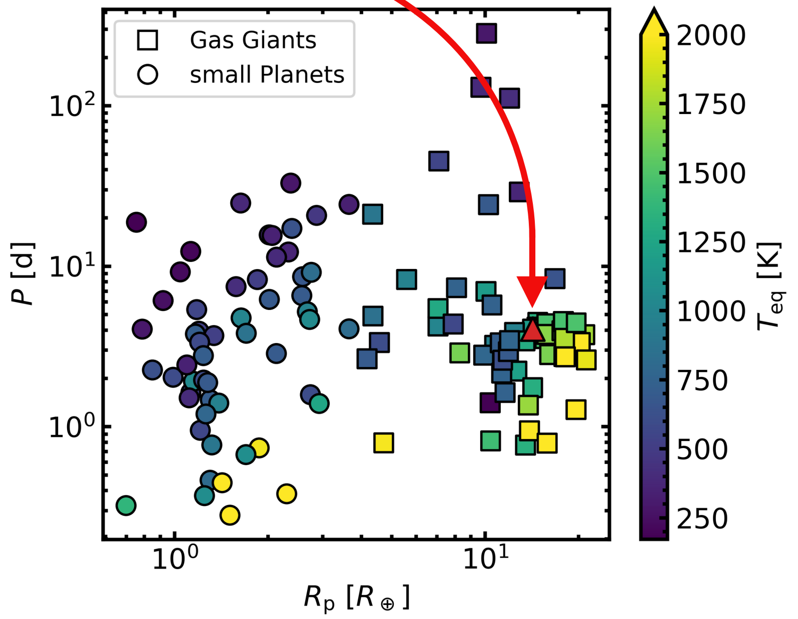

Data Reduction

Eureka! (Bell et al., 2022)

- End-to-end open source pipeline
- First part: adjusted `jwst` pipeline wrap to calibrate and reduce light-curves

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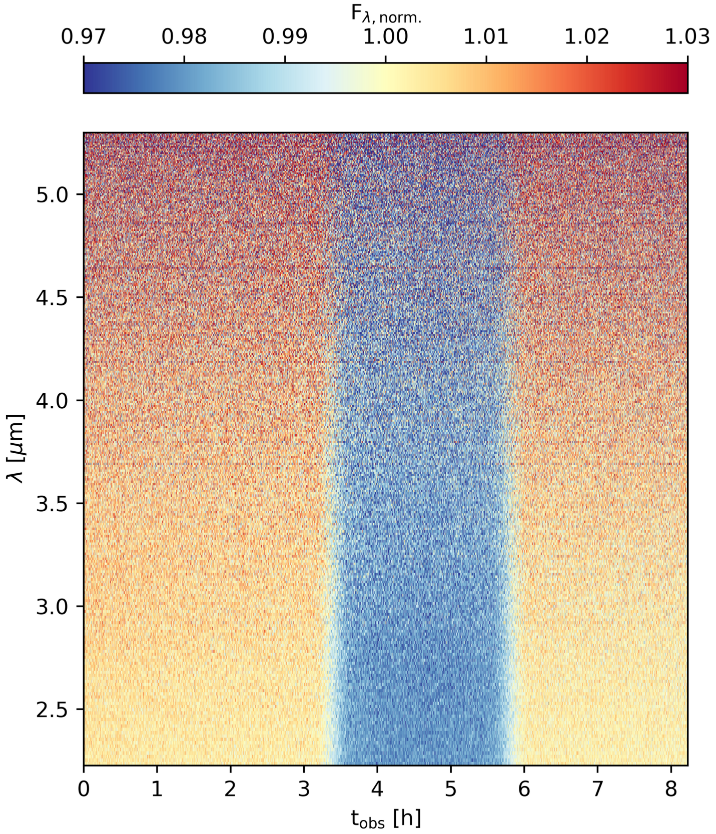

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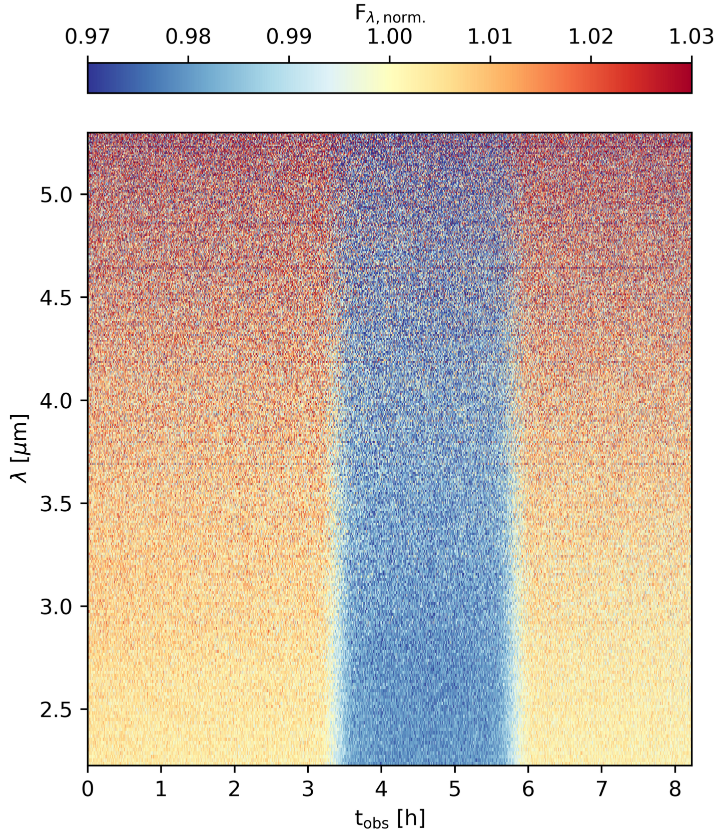
- Second part: det.-level light-curve fitting
- Determination of transit depth, $(R_p / R_s)^2$



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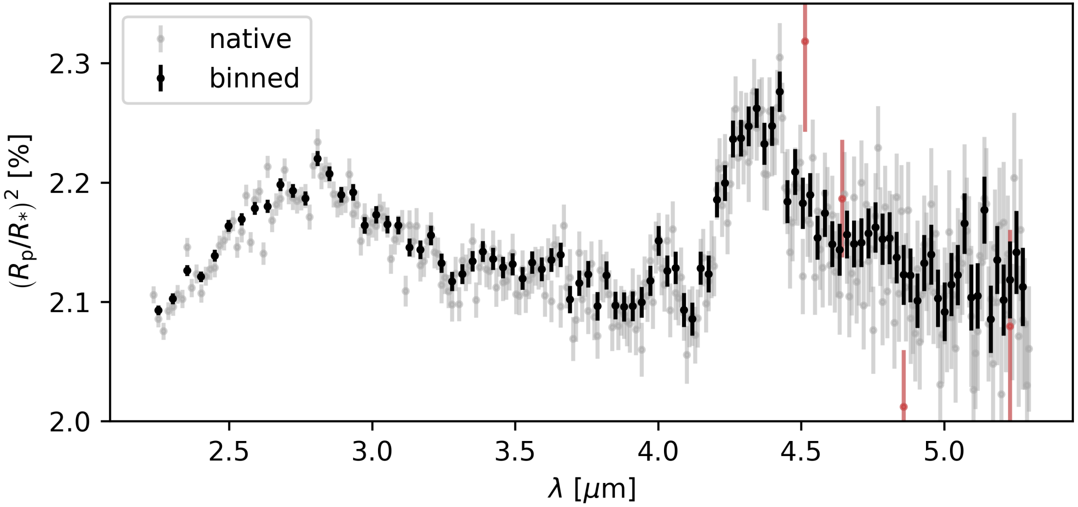

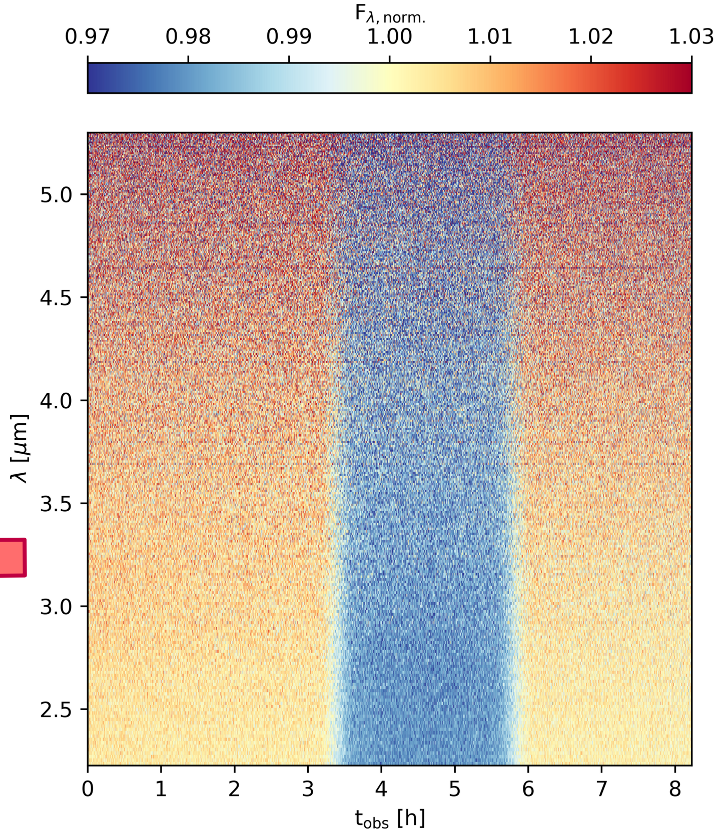

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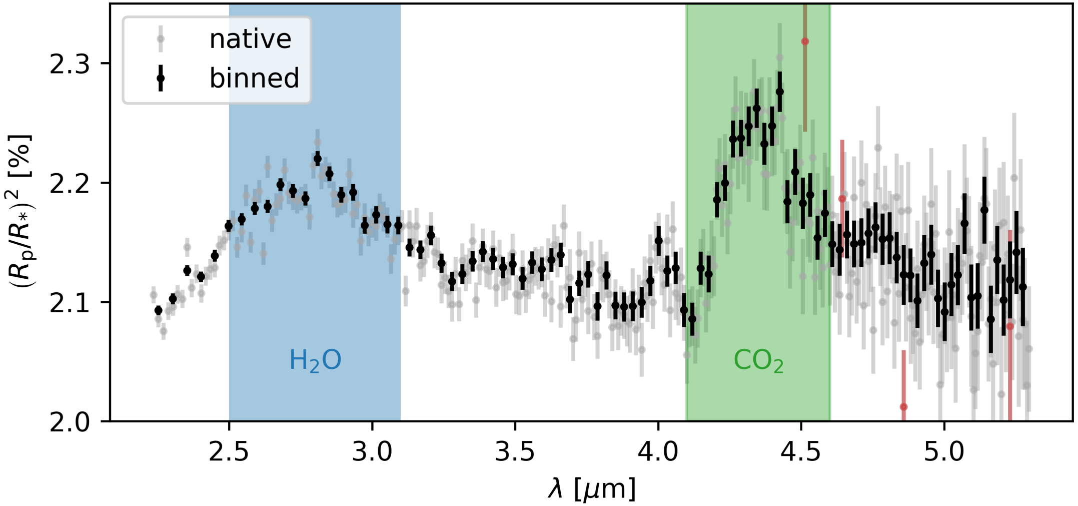
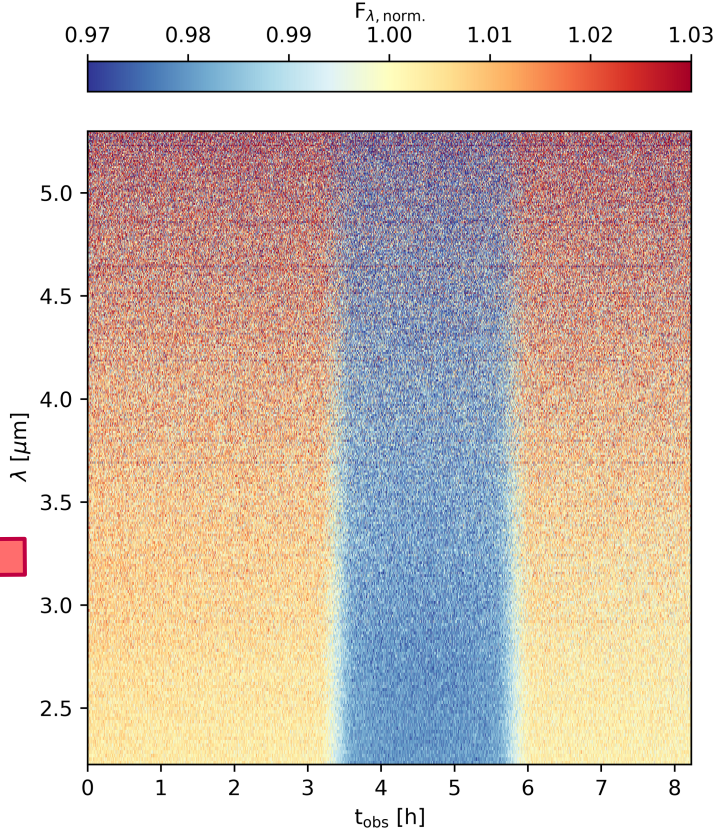
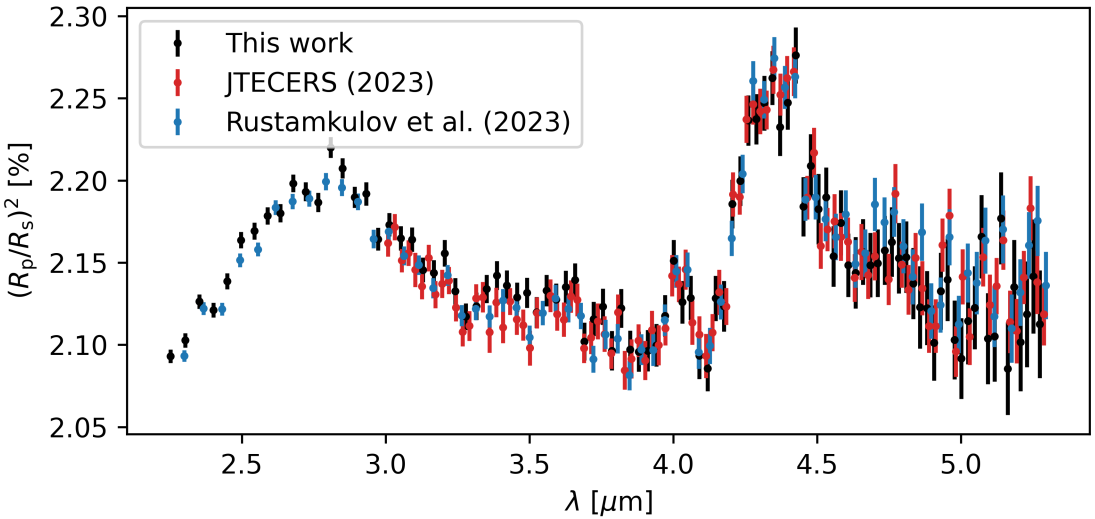




Existing Spectra

- General agreement, slight offset at ~3.5 μm
- Variation in wavelength constraints in all cases
- How does this influence characterisation results?



Atmospheric Characterisation

TauREx (Al-Refaie et al., 2021)

Primary Atmosphere (H_2 / He - dominated)

+

Minor constituents (H_2O, CO_2, CO, CH_4)

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Simple opacity-step for cloud deck

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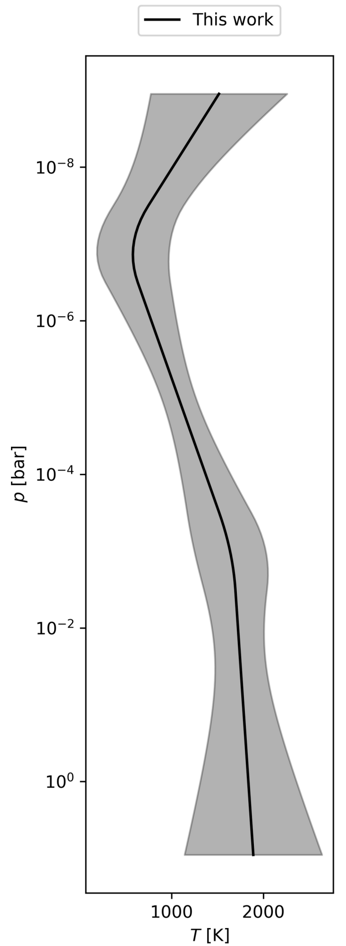
→

Bayesian inference using Nested Sampling

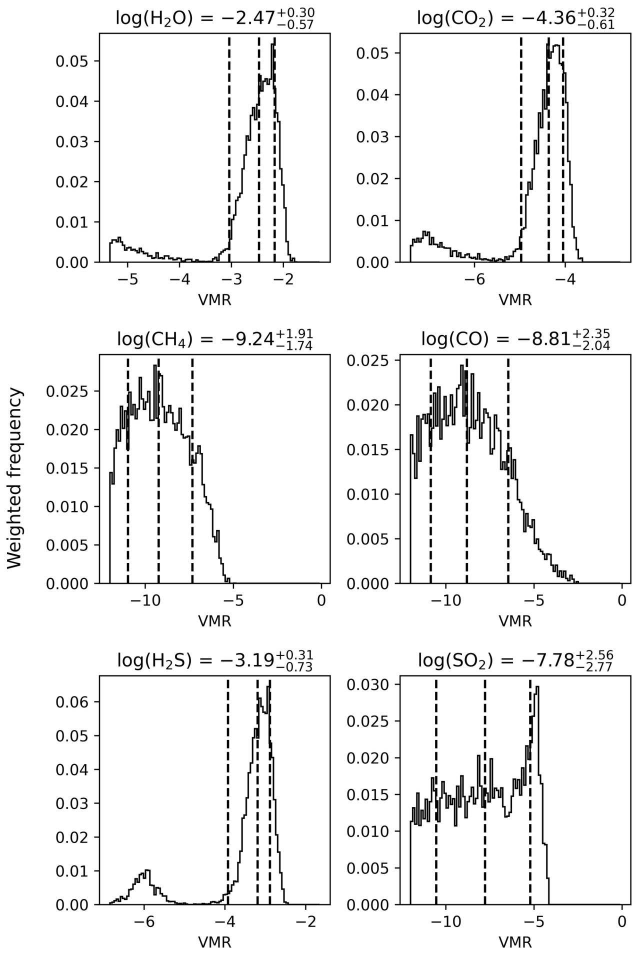
Using a 4-point p-T profile:

Determine preferred atmospheric parameters through model selection

Atmospheric Characterisation

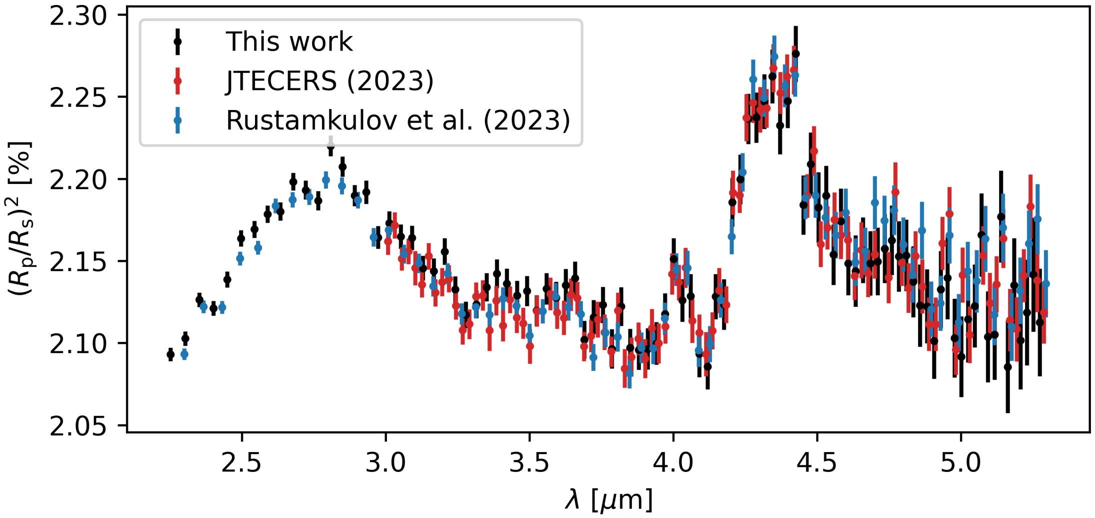


- Constraints on H_2O, CO_2, H_2S
- Upper limits on CH_4, CO, SO_2
- Small secondary solution

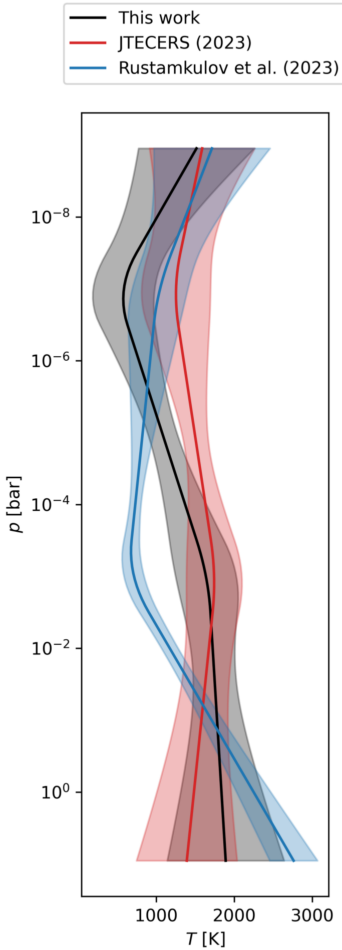


Existing Spectra

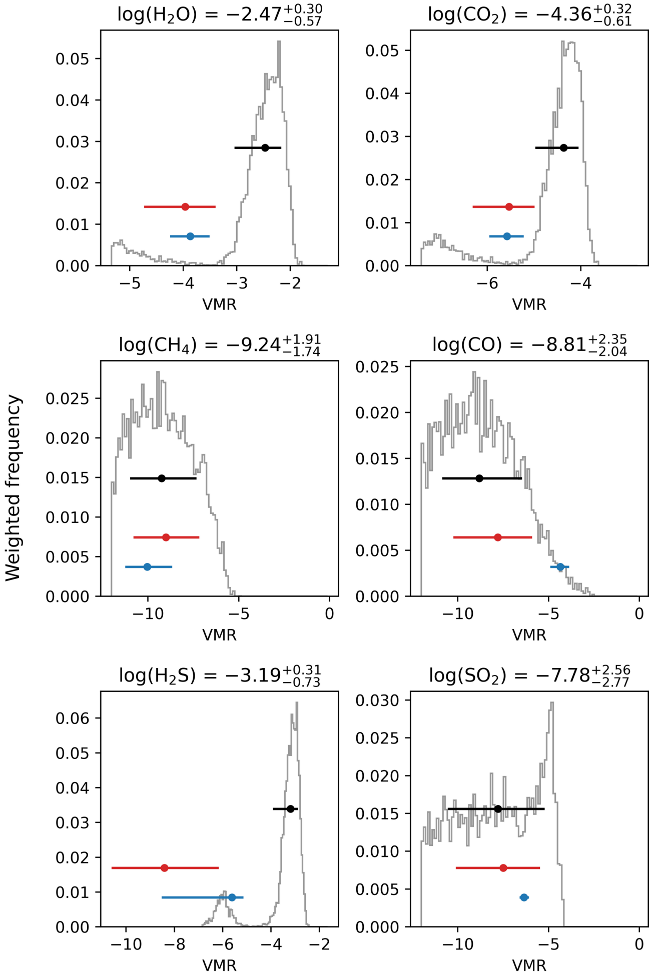
- Application of the same set-up to existing data reduction results
- Is the characterisation result consistent?



Characterisation Comparison



- Similarities in p-T profile
- Disagreement between retrieved characteristics
  - VMR constraints and values
  - Wavelength-coverage for JTECERS (2023)



Summary

- JWST will offer a large sample of exoplanet atmospheric spectra
  - Comparative planetology

- Atmospheric retrievals are very sensitive to small-scale differences between data reduction results
  - High precision of JWST transmission spectroscopy
  - Population-level analysis will require consistent data reduction

Additional Slides

References

- Al-Refaie, A. F., Changeat, Q., Waldmann, I. P., & Tinetti, G. 2021, ApJ, 917, 37
- Bell, T. J., Ahrer, E.-M., Brande, J., et al. 2022, JOSS, 7, 4503
- JWST Transiting Exoplanet Community Early Release Science Team, Ahrer, E.-M., Alderson, L., et al. 2023, Nature, 614, 649
- Rustamkulov, Z., Sing, D. K., Mukherjee, S., et al. 2023, Nature, 614, 659

Images Sources

1) https://www.nasa.gov/universe/nasas-webb-detects-carbon-dioxide-in-exoplanet-atmosphere/
2) https://www.esa.int/ESA_Multimedia/Images/2019/12/Exoplanet_phase_curve